Google Cloud

PRESENTS

Agentic AI Day

Improving Safety at Large Public Events

Powered by H2S

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Improving Safety at Large Public Events

Large-scale public events like music festivals or rallies are incredibly complex to manage. The fluid dynamics of large crowds create significant risks, from dangerous bottlenecks and crushing to difficulties in locating individuals in medical distress. Security teams need a way to move from reactive monitoring to proactive, intelligent intervention

Bengaluru Stadium Crush
M Chinnaswamy Stadium, July 2025

11 deaths
50+ injuries

2025 KUMBH MELA

80+ deaths
200+ injuries

What’s the Actual Problem??



- Reactive Decision-Making
- Lack of Real-Time Alerts
- Poor Communication
- Missing Persons Reports
- Responder Confusion

Key features

Predictive Crowd Management

Leverages Auto-ML models and movement data to forecast crowd bottlenecks in real time (5–10 min ahead).

Agentic AI Chatbot (The Brain)

A context-aware Gemini-powered chat interface that acts as the orchestrator.

Intelligent Dispatch Automation

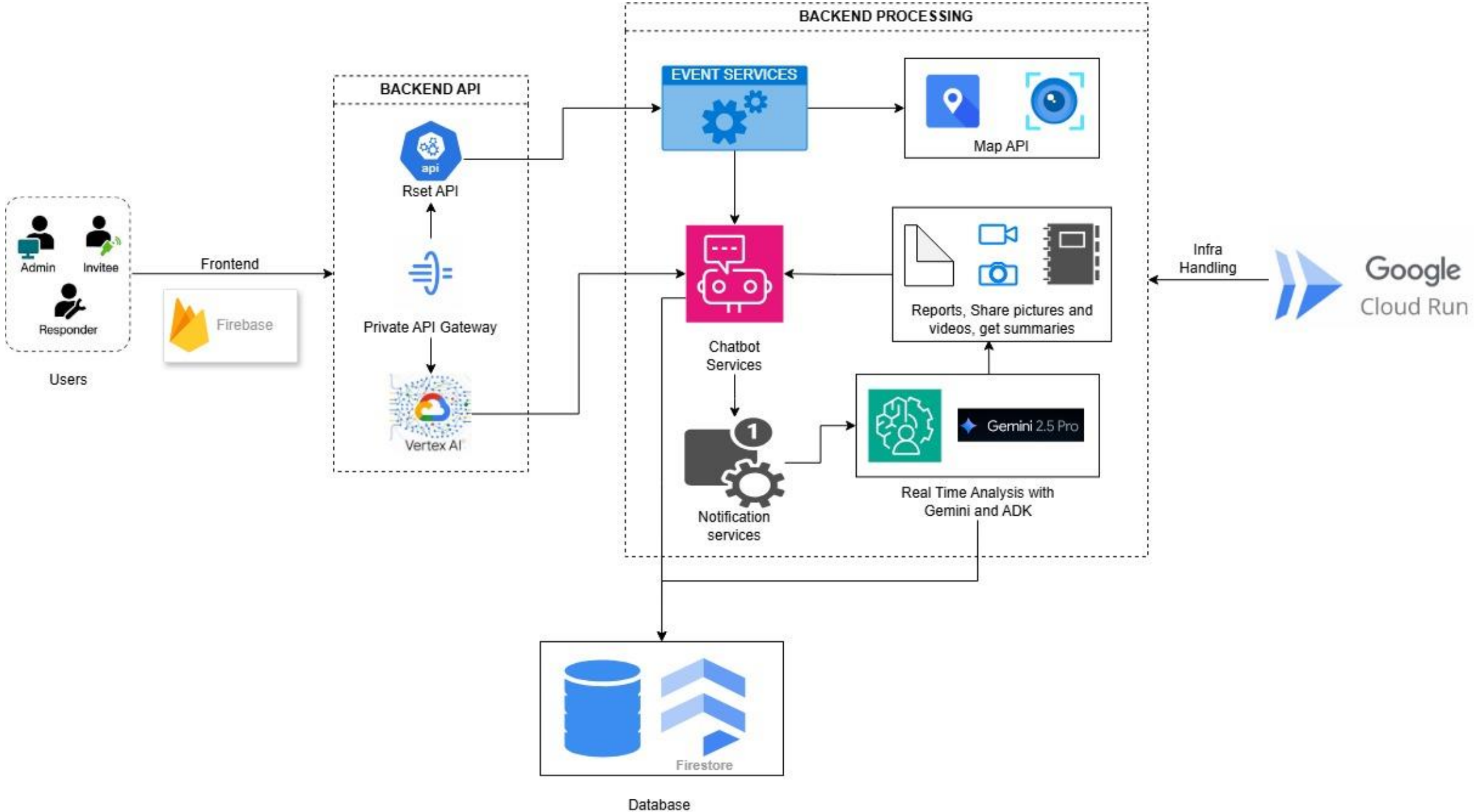
When critical zones or user/responder incident are detected or help is requested.

Lost & Found (Go Beyond)

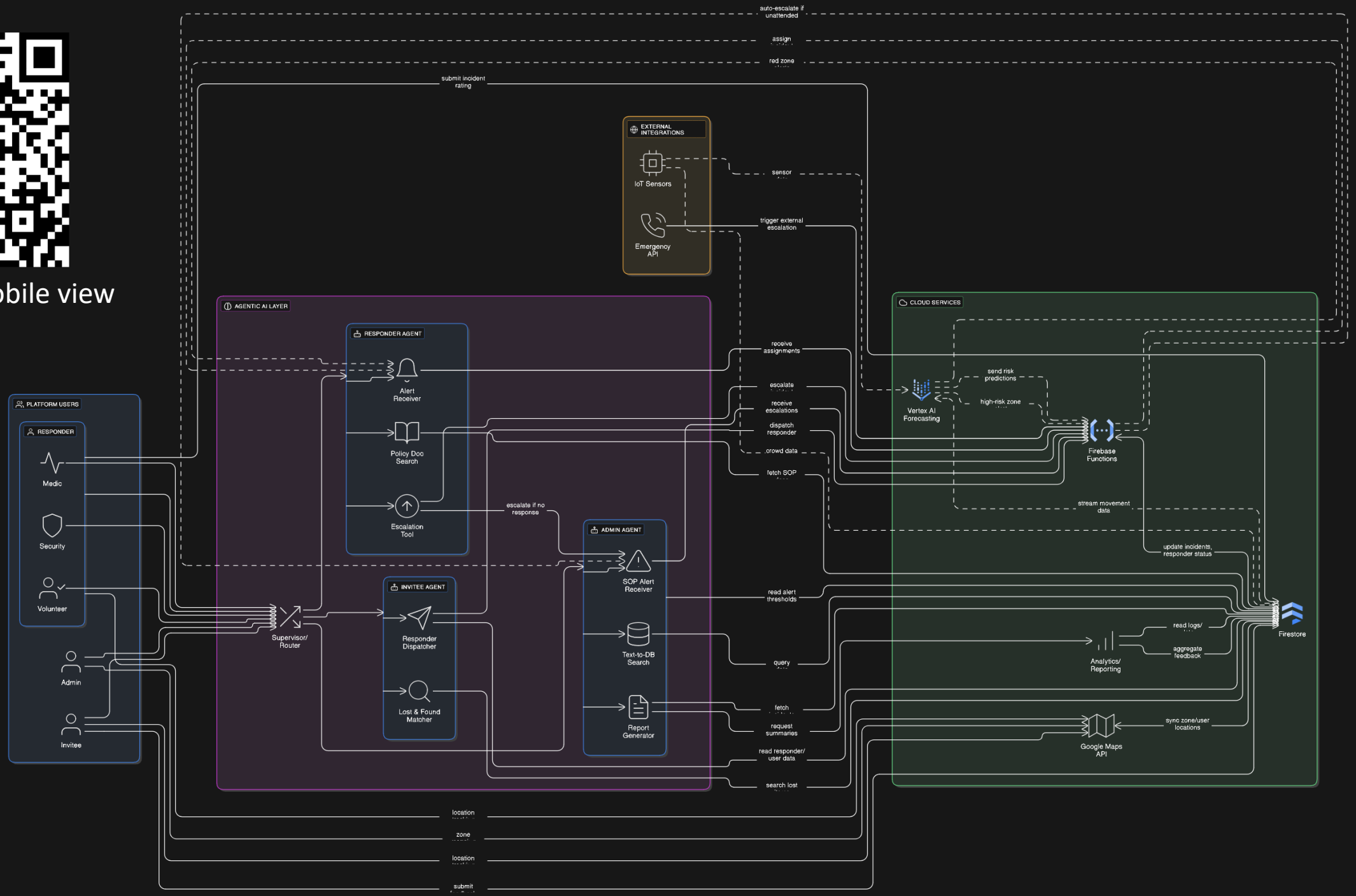
Gemini-powered matching of lost person photos with check-in/exit photos & responder data.

Multimodal Anomaly Detection

Future-scope module for detecting threats via CCTV feeds, Crowd audio patterns, Sensor inputs (motion, infrared)


BACKEND PROCESSING
BACKEND API
EVENT SERVICES
Map API
api
Rset API
Private API Gateway
Vertex AI
Admin
Invitee
Responder
Users
Frontend
Firebase
Chatbot
Services
Reports, Share pictures and
videos, get summaries
Infra
Handling
Google
Cloud Run
Gemini 2.5 Pro
Real Time Analysis with
Gemini and ADK
Notification
services
Firestore
Database

Scan to get a mobile view



Impact of the Solution

Proactive Safety

Predicts crowd surges **5–10 minutes in advance**, helping prevent stampedes and bottlenecks.

Faster Emergency Response

Automated dispatch cuts **response time by up to 60%**, ensuring help reaches the right place, fast.

Intelligent Coordination

Agentic AI eliminates confusion by giving **clear, context-aware instructions** to every responder type.

Easier to find missing people

AI-powered image matching enables **real-time tracking of missing persons**, reducing search time drastically.

Real-Time Situational Awareness

Admins get **live dashboards** and **chatbot-based summaries**, enabling smarter decisions during chaos.

Seamless Communication

Users, admins, and responders interact through a **single AI-powered interface**, replacing slow, siloed channels.

USP

□ Agentic AI at the Core

🔁 Proactive > Reactive

□ Multimodal Intelligence

📡 Dynamic Responder Routing

🗣️ Natural Language Interface

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Google Services Used

Firebase Studio
Google Cloud functions
Firebase Hosting
Google Maps API
Langchain
Gemini 2.5 Pro (Google AI)
Vertex AI Forecasting
Firebase Firestore
Firebase Functions
Firebase Cloud Messaging (FCM)

Future Scope

Watch us build the next book my show type event application, that democratizes enterprise grade security at your fingertips!